# MD. ASAD-AL-ADIL

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Rajshahi, Bangladesh

## Research Profile

Final-year Computer Science and Engineering undergraduate at **Rajshahi University of Engineering & Technology (RUET)** with a **CGPA of 3.92/4.00**, focused on **computer architecture, VLSI, and efficient AI acceleration**. My undergraduate thesis, **HeSA-TAPoT**, investigates a **boundary-aware heterogeneous systolic array architecture** for multiplierless CNN acceleration using **two-term Additive Power-of-Two (APoT) quantization**. I have hands-on experience with **Verilog RTL, STM32 embedded systems, PyTorch-based CNNs, and the Cadence VLSI flow**, with **two conference manuscripts currently under review**. I am seeking a research-oriented Master's opportunity in **energy-efficient computer architecture, VLSI, and AI accelerators**.

## Research Interests

- Computer Architecture
- AI/ML Accelerators
- Energy-Efficient & Parallel Computing
- FPGA Acceleration
- Hardware–Software Co-design
- VLSI Design

## Education

**2022–2027** **B.Sc. in Computer Science and Engineering**, Rajshahi University of Engineering & Technology (RUET), Bangladesh — **CGPA: 3.92/4.00**

**2021** Higher Secondary Certificate (HSC) — **GPA: 5.00/5.00**

**2019** Secondary School Certificate (SSC) — **GPA: 5.00/5.00**

## Undergraduate Thesis

**HeSA-TAPoT: A Boundary-Aware Heterogeneous Systolic Array Architecture for Multiplierless CNN Acceleration Using Two-Term Additive Power-of-Two Quantization** — Ongoing

- Developing a **boundary-aware heterogeneous systolic architecture** in **Verilog HDL** for efficient multi-channel CNN acceleration.
- Using **two-term APoT quantization** to replace weight multiplications with two shift operations and an addition, enabling a multiplierless convolution datapath with heterogeneous processing elements.
- Designing and verifying convolution, accumulation, pooling, bias, and requantization stages while studying throughput, resource use, numerical precision, and energy-efficiency trade-offs.

## Conference Manuscripts Under Review

**ICCIT 2026 – Manuscript on Software-Level Two-Term Shift-Based CNN Quantization** — Submitted / Under Review

Develops and evaluates a software-level **two-term shift representation** in which CNN weight multiplication is approximated using **dual shift operations plus addition**, providing the algorithmic basis for multiplierless hardware acceleration.

**ICECE 2026 (BUET) – Manuscript on Knowledge-Distilled CNN Inference on MCU** — Submitted / Under Review

Uses **knowledge distillation** to obtain a compact CNN for microcontroller inference; a preprocessed **image array is transmitted to the MCU**, which performs digit prediction with integer value only and returns a **single predicted digit**.

## Research & Technical Experience

**VLSI Design Trainee, Neural Semiconductor Limited, Dhaka** — 23 Feb 2026 – 6 Mar 2026

- Gained hands-on exposure to the digital VLSI design flow using **Cadence**, including RTL design, functional verification, physical-layout concepts, and fabrication-oriented constraints.
- Studied **DRC/LVS** verification and the transition from digital design to silicon implementation.

## Selected Engineering Projects

**Handwritten Digit Recognition using STM32 and UART** GitHub

- Developed a PyTorch CNN for handwritten-digit recognition using **standard and custom datasets**, achieving **98.2% classification accuracy**.
- Built a Python-based handwritten-input acquisition and preprocessing pipeline and transmitted the **predicted digit through UART** to an **STM32 microcontroller**.
- Programmed the STM32 in embedded C to receive the classification result and display it on a **7-segment display** through custom digital interfacing.

**Medical Charges Predictor using Machine Learning** Hugging Face

Built and deployed a Gradient-Boosting model for healthcare-cost estimation using preprocessing, feature engineering, hyperparameter tuning, cross-validation, and an interactive inference interface.

**Additional Software Projects**

**MyExpense**: Django/SQLite personal-finance web application (GitHub);
**Expense Tracker**: Java/Android application for persistent expense recording and monthly visualization (GitHub).

## Technical Skills

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|---|---|
| **HDL / Architecture:** | Verilog HDL, RTL Design, Digital Logic, Systolic / PE-based Architecture |
| **Programming:** | C, C++, Python, Java |
| **VLSI / EDA:** | Cadence, Logisim Evolution |
| **Embedded Systems:** | STM32F103C8, Embedded/Bare-Metal C, UART, Arduino, Proteus, Keil $\mu$Vision |
| **AI / ML:** | PyTorch, Knowledge Distillation, Model Compression, Scikit-learn, NumPy, Pandas |
| **Development:** | Git/GitHub, VS Code, Matplotlib, Plotly |

## Honors & Leadership

**Student of the Year Award – 2nd Academic Year** 2023
**Class Representative Appreciation – 3rd Academic Year** 2024

## Additional Experience

**Academic Tutor & Script Evaluator, Private Educational Organization** Sep 2022 – Present

- Support academic instruction and have evaluated more than **13,000 school-level examination scripts** using structured rubrics and written performance feedback.
- Provide **one-to-one private tutoring and personalized academic support** in Physics, Mathematics, and IT-related subjects for students requiring individualized guidance.

## Interests

**Strength Training & Fitness:** Maintaining a consistent strength-training routine for approximately six years; placed **5th in the Mr. Rajshahi Competition** in 2024.

## Academic References

**Dr. Md. Nazrul Islam Mondal** (Supervisor)
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*Currently on study leave for PhD*